Dynamic Response 1

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Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

Content

1. Response by convolution
2. Transfer Functions and Frequency Response

1. Response by convolution

- Two attributes of linear time-invariant systems (LTIs):
  - The response obeys the principle of superposition.
  - The response can be expressed as the **convolution** of the **input** with the **unit impulse response** of the system.

Superposition

- Applying the input $u_1(t)$ => output $y_1(t)$
- Applying the input $u_2(t)$ => output $y_2(t)$
- Applying the input $u(t)= \alpha_1 u_1(t) + \alpha_2 u_2(t)$ =>

  output $y(t)= \alpha_1 y_1(t) + \alpha_2 y_2(t)$.

1. Response by convolution

Superposition

- Applying the input u1(t) => output y1(t)
- Applying the input u2(t) => output y2(t)
- Applying the input u(t)= α1u1(t) + α2u2(t) => output y(t)= α1y1(t) + α2y2(t).

$$\dot{y} + ky = u.$$

$$u = \alpha_1 u_1 + \alpha_2 u_2$$

$$y = \alpha_1 y_1 + \alpha_2 y_2$$

$$\alpha_1 \dot{y}_1 + \alpha_2 \dot{y}_2 + k(\alpha_1 y_1 + \alpha_2 y_2) = \alpha_1 u_1 + \alpha_2 u_2$$

$$\alpha_1(\dot{y}_1 + ky_1 - u_1) + \alpha_2(\dot{y}_2 + ky_2 - u_2) = 0.$$

If y1 is the solution with input u1
And
If y2 is the solution with input u2
then the equation is satisfied

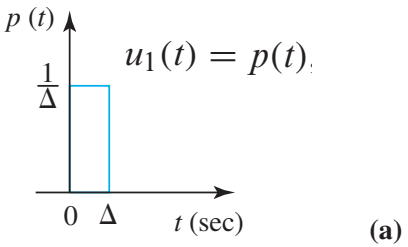
1. Response by convolution

Superposition

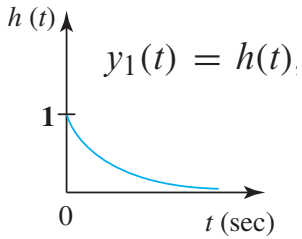
- We are able to solve for the response of a linear system to a general signal simply by:
  - decomposing the given signal into a sum of the elementary components
  - by superposition, the general signal is the sum of the responses to the elementary signals
- The most common candidates for elementary signals in LTI:
  - Impulse
  - Exponential

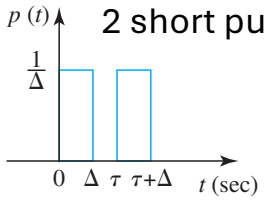
1. Response by convolution

Superposition

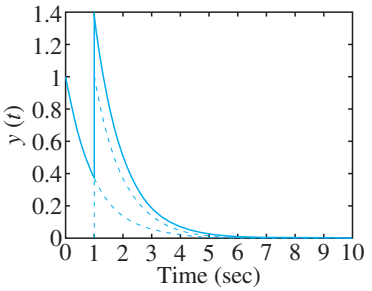


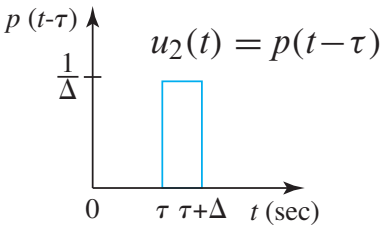
(a)



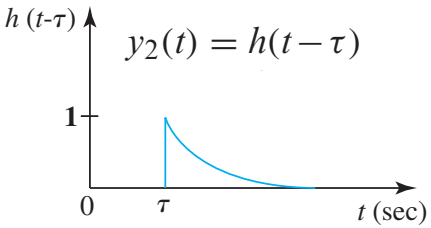


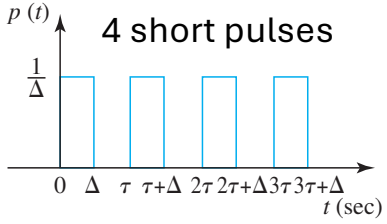
(c)



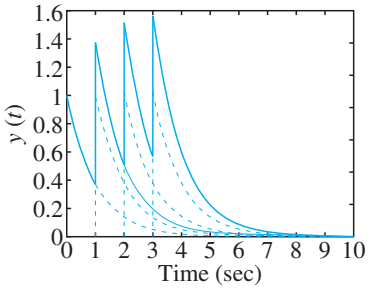


(b)





(d)



1. Response by convolution

2 short pulses

Superposition

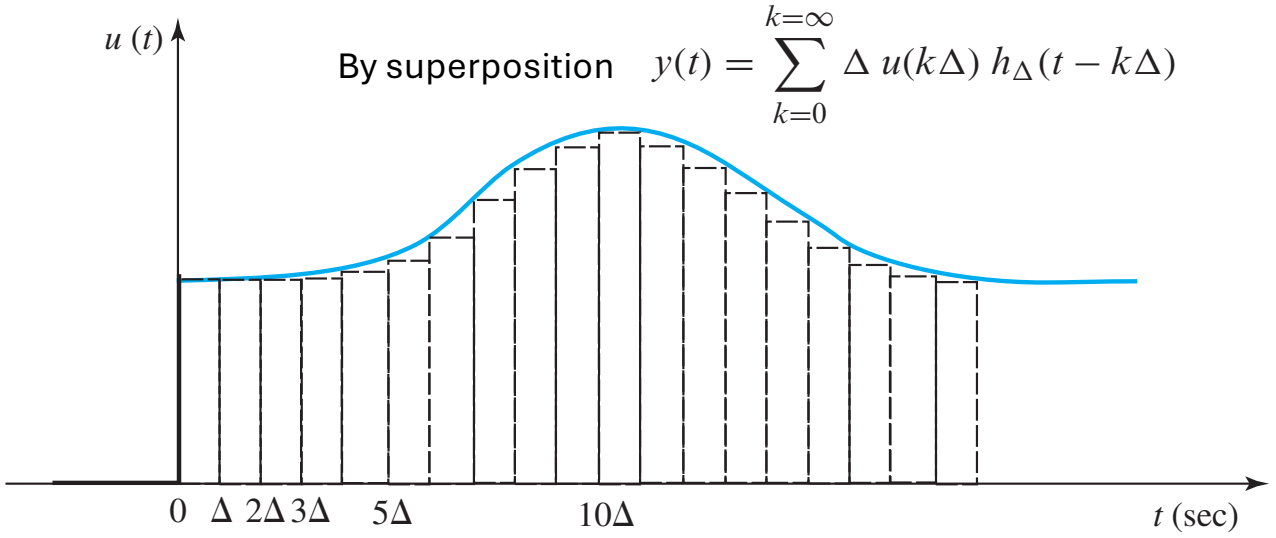
- Any arbitrary input signal u(t) may be approximated by series of pulses
- Rectangular pulse with **unit area** $p_\Delta(t)$

$$p_\Delta(t) = \begin{cases} \frac{1}{\Delta}, & 0 \le t \le \Delta \\ 0, & \text{elsewhere} \end{cases}$$

Response to $p_\Delta(t)$ => $h_\Delta(t)$

Response at time $n\Delta$ to $\Delta\, u(k\Delta) p_\Delta(k\Delta)$ => $\Delta\, u(k\Delta)\, h_\Delta(n\Delta - k\Delta)$

By superposition $y(t) = \sum_{k=0}^{k=\infty} \Delta\, u(k\Delta)\, h_\Delta(t - k\Delta)$



1. Response by convolution

Convolution

Δ -> 0, then basic pulse gets more and more narrow and taller (holding a constant area)

- Impulse signal δ(t) we can treat continuous signals

$$\Delta \to 0 \qquad \lim_{\Delta \to 0} p_\Delta(t) = \delta(t) \qquad \lim_{\Delta \to 0} h_\Delta(t) = h(t) = \text{the impulse response.}$$

$$y(t) = \int_0^\infty u(\tau) h(t-\tau)\, d\tau$$

Changing the variable $\tau_1 = t - \tau$

Convolution integral => $$y(t) = \int_\infty^{-\infty} u(t-\tau_1) h(\tau_1)\,(-d\tau_1) = \int_{-\infty}^{\infty} h(\tau) u(t-\tau)\, d\tau$$

Output y(t) for a general input u(t - τ)

1. Response by convolution

Convolution

- For time-invariant systems, the output for a general input is given by the integral

$$y(t) = \int_{-\infty}^{\infty} u(\tau)h(t-\tau)d\tau$$

- For causal systems (the output is a reaction of the input or y(t) = 0 if u(t) = 0) and considering input starts at t = 0

$$y(t) = \int_{0}^{t} u(\tau)h(t-\tau)d\tau.$$

1. Response by convolution

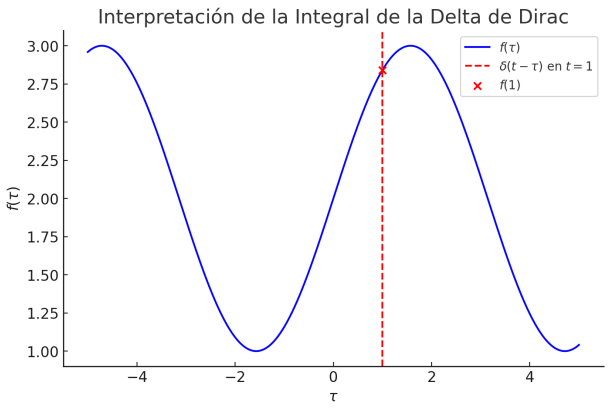
- Definition of impulse (E.g. motion of a baseball hit by a bat):
  - Analysis is complex (deformation of the ball and the bat)
  - For calculating the ball's path => collision = **net velocity change** of the ball over a **very short time** period (**impulse**)

$$\delta(t) = 0 \quad t \neq 0,$$

$$\int_{-\infty}^{\infty} \delta(t)dt = 1.$$

$$\int_{-\infty}^{\infty} f(\tau)\delta(t-\tau)\,d\tau = f(t)$$

Sifting property of impulse



1. Response by convolution

Example (convolution): Determine the impulse response for the system described by the differential equation with an initial condition of y(0) = 0 before the impulse.

$$\dot{y} + ky = u = \delta(t),$$

Given the impulse we can integrate

$$\int_{0^-}^{0^+} \dot{y}\,dt + k\int_{0^-}^{0^+} y\,dt = \int_{0^-}^{0^+} \delta(t)\,dt.$$

$y(0^+) - y(0^-)$; Aprox. 0 ; 1

$$y(0^+) - y(0^-) = 1.$$

With $y(0^-) = 0$, Then $y(0^+) = 1$

For t > 0 (after the impulse)

$$\dot{y} + ky = 0, \qquad y(0^+) = 1.$$

Assuming the solution $y = Ae^{st}, \quad \dot{y} = Ase^{st}$

$$Ase^{st} + kAe^{st} = 0,$$

$$s + k = 0,$$

With $y(0^+) = 1 \Rightarrow A = 1$

$$s = -k.$$

$$y(t) = h(t) = e^{-kt} \text{ for } t > 0.$$

1. Response by convolution

Example (convolution): Determine the impulse response for the system described by the differential equation with an initial condition of y(0) = 0 before the impulse.

$$y(t) = h(t) = e^{-kt} \text{ for } t > 0.$$

or

$$h(t) = e^{-kt}1(t)$$

Impulse response of the first-order system

with

$$1(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

The response of this system to a general input: convolution of this impulse response with the input

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau)u(t-\tau)\,d\tau \\ &= \int_{-\infty}^{\infty} e^{-k\tau}1(\tau)u(t-\tau)\,d\tau \\ &= \int_{0}^{\infty} e^{-k\tau}u(t-\tau)\,d\tau. \end{aligned}$$

2. Transfer Functions and Frequency Response

- Finding the response with the convolution may be difficult in some cases
- The process is simpler by applying the Laplace transform

Laplace transform

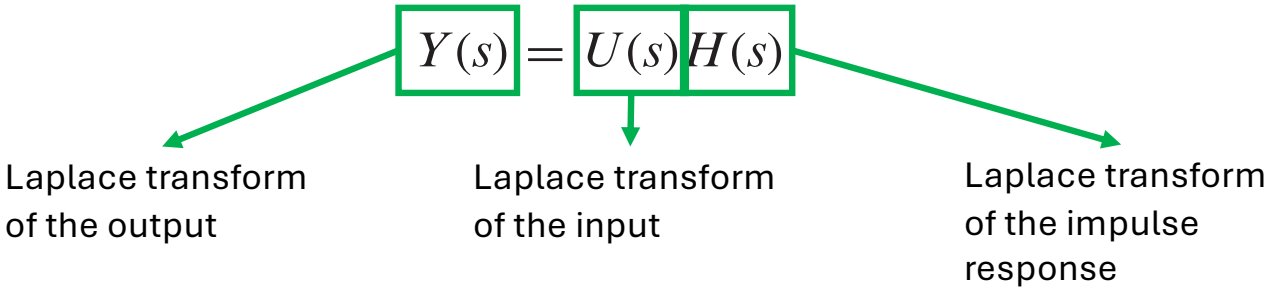
$$Y(s) = \int_{-\infty}^{\infty} y(t)e^{-st}dt.$$

Applying Laplace transform to the convolution

$$Y(s) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\tau)u(t-\tau)d\tau \right] e^{-st}dt$$

$$Y(s) = U(s)H(s)$$

2. Transfer Functions and Frequency Response



Transfer function

- The integrals of the transforms usually do not converge for all values of “s”
- The integrals are only defined for a finite region in the s-plane.
- “s” may be complex ($s = \sigma_1 + j\omega$) => input and output may be complex

2. Transfer Functions and Frequency Response

$$Y(s) = U(s)H(s)$$

$$Y(s) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\tau)u(t-\tau)d\tau \right] e^{-st}dt$$

If the input u(t) has the form e^{st}

The output y(t) has the form $H(s)e^{st}$

"s" can be complex: $s = \sigma_1 + j\omega$

The output can be complex

$$y(t) = \int_{-\infty}^{\infty} h(\tau)u(t-\tau)\,d\tau$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}\,d\tau,$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{st}e^{-s\tau}\,d\tau,$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}\,d\tau e^{st},$$

$$y(t) = H(s)e^{st},$$

Where: $H(s) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}\,d\tau.$[3]

2. Transfer Functions and Frequency Response

- Limits on the integral: −∞ to +∞ => h(t) may have values at **any time** (−∞ < t < ∞)
- **No initial conditions**
- If the input is an exponential time function => so is the output and the scaling term is the transfer function
- For any real, causal system, h(t) = 0 for t < 0

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau}\, d\tau.$$

$$H(s) = \int_{0}^{\infty} h(\tau) e^{-s\tau}\, d\tau$$

$$y(t) = \int_{0}^{\infty} h(\tau) u(t-\tau)\, d\tau.$$

2. Transfer Functions and Frequency Response

Example (transfer function): Compute the transfer function for the system of the previous example, and find the output **y** for all time (**−∞ < t < ∞**) when the input **u = exp(st)** for all time and **s** is a given complex number.

$$\dot{y}(t) + ky(t) = u(t) = e^{st}.$$

$$y(t) \text{ as } H(s)e^{st}$$

$$sH(s)e^{st} + kH(s)e^{st} = e^{st}.$$

$$H(s) = \frac{1}{s+k}.$$

$$y(t) = \frac{e^{st}}{s+k}.$$

2. Transfer Functions and Frequency Response

Definitions (2 options **assuming** that **all the initial conditions** are **zero**):

- The transfer function H(s) is the ratio of the Laplace transform of the output of the system to its input assuming all zero initial conditions.

$$\frac{Y(s)}{U(s)} = H(s)$$

- The transfer function H(s) is the Laplace transform of the unit impulse response h(t) (U(s) = 1 or u(t) = δ(t)).

$$Y(s) = H(s)$$

2. Transfer Functions and Frequency Response

- To characterize an LTI system:
  - Applying a unit impulse
  - Measure the response

$$Y(s) = H(s)$$

- For a system described by an ordinary differential equation

$$\dddot{y} + a_1\ddot{y} + a_2\dot{y} + a_3 y = b_1\ddot{u} + b_2\dot{u} + b_3 u,$$

$$(s^3 + a_1 s^2 + a_2 s + a_3)\,Y(s) = (b_1 s^2 + b_2 s + b_3)\,U(s)$$

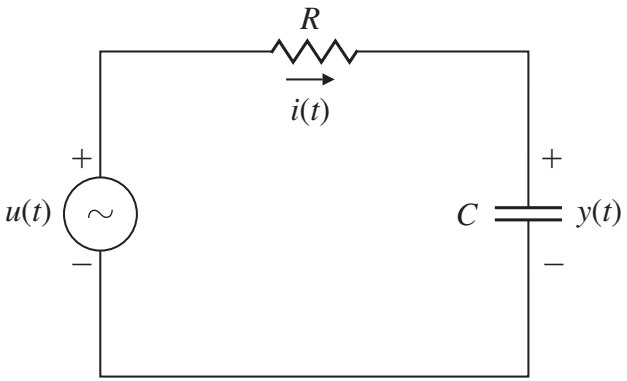
$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3} = \frac{b(s)}{a(s)}$$

Assuming zero initial conditions

$$s^3\,Y(s) + a_1 s^2\,Y(s) + a_2 s\,Y(s) + a_3\,Y(s) = b_1 s^2\,U(s) + b_2 s\,U(s) + b_3\,U(s)$$

2. Transfer Functions and Frequency Response

Example: Transfer Function for an RC Circuit



$$Ri(t) + y(t) = u(t),$$

$$i(t) = C\frac{dy(t)}{dt}$$

$$RC\dot{y} + y = u(t).$$

If the input voltage is a unit impulse signal

$$RC\dot{y} + y = \delta(t)$$

$$RC(sY(s) - y(0^-)) + Y(s) = U(s) = 1$$

$$H(s) = \frac{Y(s)}{U(s)} = Y(s) = \frac{1}{RCs + 1}$$

$$y(t) = h(t) = \frac{1}{RC}e^{-\frac{t}{RC}}1(t)$$

$$H(s) = \mathcal{L}\{h(t)\} = \frac{1}{RCs + 1}$$

2. Transfer Functions and Frequency Response

Frequency response: response to a sinusoid

Analysis by using the response to an exponential input

Response for an exponential input $y(t) = H(s)e^{st}$

Sinusoidal input $A\cos(\omega t) = \frac{A}{2}(e^{j\omega t} + e^{-j\omega t})$

$$s = j\omega$$

$$u(t) = e^{j\omega t} \quad y(t) = H(j\omega)e^{j\omega t}$$

$$u(t) = e^{-j\omega t} \quad y(t) = H(-j\omega)e^{-j\omega t}$$

By superposition $y(t) = \frac{A}{2}\left[H(j\omega)e^{j\omega t} + H(-j\omega)e^{-j\omega t}\right]$

The transfer function H(jω) is a complex number

$$H(j\omega) = M(\omega)e^{j\varphi(\omega)}$$

$$H = Me^{j\varphi}$$

$$M = |H(j\omega)|,\ \varphi = \angle H(j\omega).$$

Input: cos( ) with magnitude A

Output: cos ( ) with magnitude AM and shifted an angle ϕ

$$y(t) = \frac{A}{2}M\left(e^{j(\omega t+\varphi)} + e^{-j(\omega t+\varphi)}\right)$$

$$= AM\cos(\omega t + \varphi),$$

2. Transfer Functions and Frequency Response

Frequency response example: 1) find the frequency response to a sinusoidal input u = A cos(ωt).

$$\dot{y} + ky = u.$$

$$k = 1.$$

$$u = A\cos(\omega t)$$

$$H(s) = \frac{1}{s+k} \Longrightarrow H(j\omega) = \frac{1}{j\omega + k}$$

$$M = \left|\frac{1}{j\omega + k}\right| = \frac{1}{\sqrt{\omega^2 + k^2}}$$

$$\varphi = -\tan^{-1}\left(\frac{\omega}{k}\right)$$

$$y(t) = AM\cos(\omega t + \varphi)$$

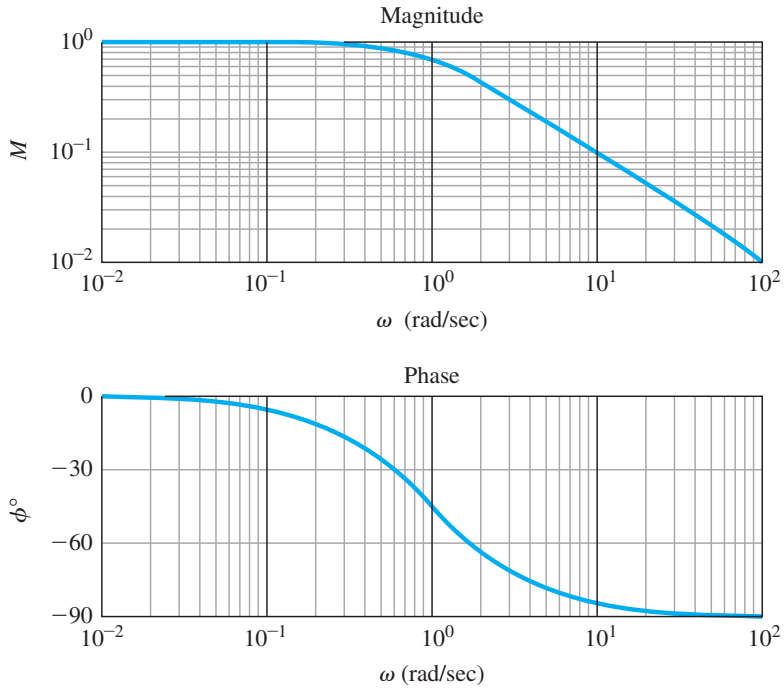
M: amplitude
Φ: phase
Both functions of ω

2. Transfer Functions and Frequency Response

Frequency response example: 1) find the frequency response to a sinusoidal input u = A cos(ωt).

```
k=1;
s=tf('s');
sysH = 1/(s+k);
w = logspace(-2,2);


[mag,phase] = bode(sysH,w);
figure; loglog(w,squeeze(mag));
figure;
semilogx(w,squeeze(phase));
```



2. Transfer Functions and Frequency Response

Procedure for determining y(t) using the Laplace transform

1. STEP 1. Determine the transfer function: H(s)= L{impulse response of the system}
   (a) Take the Laplace transform of the equations of motion.
   (b) Solve the resulting algebraic equations (block diagram or Matlab)
2. STEP 2. Determine the Laplace transform of the input signal: U(s)= L{u(t)}
3. STEP 3. Compute the Laplace transform of the output: Y (s)= H(s) U(s)
4. STEP 4. Break up Y (s) by partial-fraction expansion
5. STEP 5. Find the output of the system by computing the inverse Laplace transform of Y (s) in Step 4, $y(t)= L^{-1}\{Y (s)\}$
   (a) Look up the components of y(t) in a table of transform–time function pairs
   (b) Combine the components to give the total solution

2. Transfer Functions and Frequency Response

Frequency response
example: 2) find the response at t = 0 to u(t)= sin(10t)1(t).

$$\dot{y} + ky = u.$$

$$k = 1.$$

$$\mathcal{L}\{u(t)\} = \mathcal{L}\{\sin(10t)\} = \frac{10}{s^2 + 100}$$

$$\begin{aligned} Y(s) &= H(s)U(s) \\ &= \frac{1}{s+1}\frac{10}{s^2+100}, \\ &= \frac{\alpha_1}{s+1} + \frac{\alpha_0}{s+j10} + \frac{\alpha_0^*}{s-j10}, \\ &= \frac{\frac{10}{101}}{s+1} + \frac{\frac{j}{2(1-j10)}}{s+j10} + \frac{\frac{-j}{2(1+j10)}}{s-j10} \end{aligned}$$

$$\begin{aligned} y(t) &= \frac{10}{101}e^{-t} + \frac{1}{\sqrt{101}}\sin(10t+\varphi) \\ &= y_1(t) + y_2(t), \\ \varphi &= \tan^{-1}(-10) = -84.2^\circ. \end{aligned}$$

2. Transfer Functions and Frequency Response

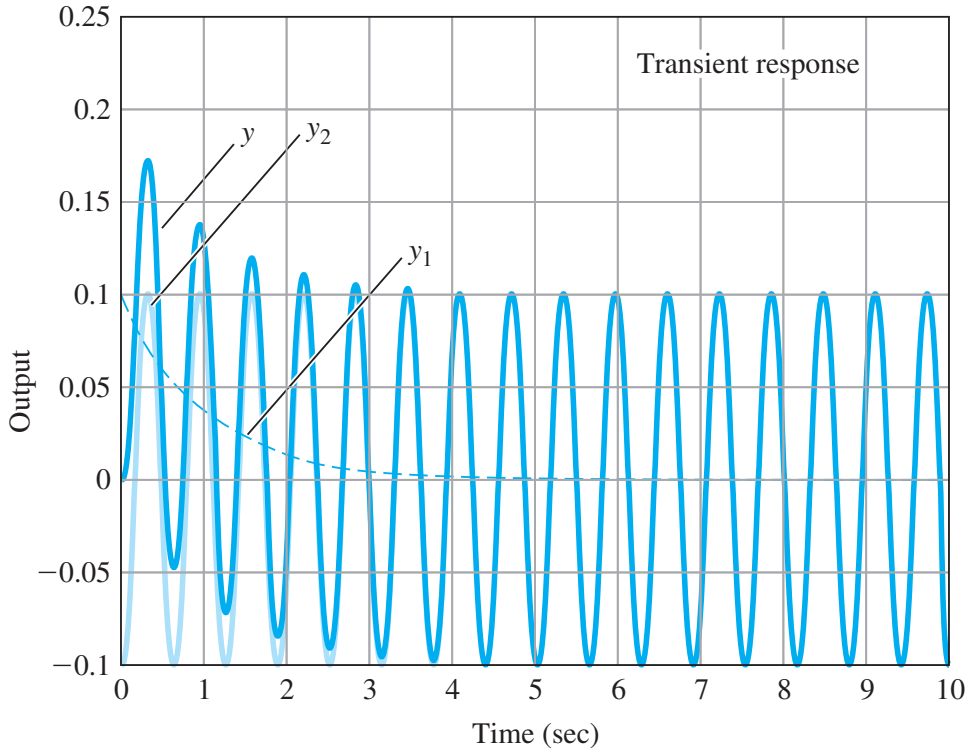
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$$k = 1.$$

$$y(t) = \frac{10}{101}e^{-t} + \frac{1}{\sqrt{101}}\sin(10t + \varphi)$$
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2. Transfer Functions and Frequency Response

Frequency response example: 2) find the response at t = 0 to u(t)= sin(10t)1(t).

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